

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO1 Statement	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name	CH.V.N.V.S SRIRAM	Reg. No.	192511011
Section/Batch	Slot C /2025	Date	03-10-2026

Code of Conduct:

I, CH.V.N.V.S SRIRAM , certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ch. Srujan

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

Q4 A ride-sharing application must efficiently match users with nearby drivers in real time. Construct a concept map connecting Algorithm → Input Size → Time Complexity → Space Complexity → Efficiency → Performance Measurement. Illustrate how increasing input size impacts system performance and establish relationships between these concepts. Analyze how the choice of algorithm affects efficiency and justify the most suitable approach.

Concept Mapping

